Daniel A. Yahalomi

MIT Kavli Institute for Astrophysics and Space Research, 70 Vassar St, Cambridge, MA 02139
yahalomi@mit.edu | danielyahalomi.com | dyahalomi | 0000-0003-4755-584X

Appointments

Massachusetts Institute of Technology — *Cambridge, MA*
Juan Carlos Torres Postdoctoral Fellow — *2026 – present*

Flatiron Institute, Center for Computational Astrophysics — *New York, NY*
Flatiron Research Fellow — *2025*

Center for Astrophysics | Harvard & Smithsonian — *Cambridge, MA*
Astronomer on TESS Science Team — *2018-2020*

Education

Columbia University — *New York, NY*
Ph.D. (2025), M.Phil. (2023), M.A. (2022) in Astronomy and Astrophysics — *2020 – 2025*
– **Advisor**: David Kipping
– **Thesis**: From Wobbles to Worlds: Developing a Framework for Detecting Unseen Planets and Moons

Massachusetts Institute of Technology — *Cambridge, MA*
B.S. in Physics; Minors in Computer Science and Comparative Media Studies — *2014 – 2018*
– **Advisor**: Paul L. Schechter
– **Thesis**: Statistical Analyses of Gravitational Microlensing Probability Densities

Fellowships, Grants, & Awards

- **MIT Juan Carlos Torres Fellowship**: 2025.
- **Flatiron Research Fellowship**: 2025.
- **Pinkerton Foundation Grant Winner ($70K)** – Columbia STAR Program: 2026.
- **Pinkerton Foundation Grant Winner ($55K)** – Columbia STAR Program: 2025.
- **Pinkerton Foundation Discretionary Fund Winner ($5K)** – Columbia STAR Program: 2024.
- **Columbia Incite Breakdown/(Re)generate Finalist ($40K)** – Columbia STAR Program: 2024.
- **NASA ExoExplorer**: 2023-2024.
- **LSST-DA DSFP Fellow**: 2023-2025.
- **AAS NOLP Fellow**: 2023-2025.
- **Dean's Fellowship**, Columbia Graduate School of Arts & Science: 2020 – 2025.
- **NSF GRFP Honorable Mention**: 2020.

Publications

Citations: 1,539 total. *h-index:* 18. *i10-index:* 24. Updated November 2026.

First Author Publications

11. **Yahalomi, D. A.** et al. "Preparing for the *Gaia* DR4 Epochalypse: Simulated Epoch Astrometry for Substellar Companion Orbit-Modeling Benchmarks." *Submitted; arXiv:2611.06915* (2026).
10. **Yahalomi, D. A.** et al. "Singles in Disguise: *Kepler*'s Low-SNR Single-Transiting Systems as Incompletely Detected Multis." *Submitted; arXiv:2610.22349* (2026).

9. **Yahalomi, D. A.** et al. "Micrometeoroid Impact Rate Analysis for an Artemis-Era Lunar Base." *In preparation; arXiv:2511.04740* (2026).
8. **Yahalomi, D. A.** et al. "The Astrometric Resoeccentric Degeneracy: Eccentric Single Planets Mimic 2:1 Resonant Planet Pairs in Astrometry." ApJL, 999, L9 (2026).
7. **Yahalomi, D. A.** et al. "The `democratic detrender`: Ensemble-Based Removal of Nuisance Signal in Stellar Time-Series Photometry." ApJS, 283, 2 (2026).
6. **Yahalomi, D. A.** and Kipping, D. "Mapping the Orbital Landscape for Perturbing Planet Solutions for Single-Planet Systems with TTVs." ApJ, 998, 136 (2026).
5. **Yahalomi, D. A.** et al. "The Exoplanet Edge: Planets Don't Induce Observable TTVs Faster than Half their Orbital Period." ApJL, 984, L67 (2025).
4. **Yahalomi, D. A.** et al. "Not So Fast Kepler-1513: A Perturbing Planetary Interloper in the Exomoon Corridor." MNRAS, 527, 620 (2024).
3. **Yahalomi, D. A.** et al. "Detecting Solar System Analogs through Joint Radial Velocity/Astrometric Surveys." AJ, 166, 258 (2023).
2. **Yahalomi, D. A.** et al. "The Mass of the White Dwarf Companion in the Self-Lensing Binary KOI-3278: Einstein vs. Newton." ApJ, 880, 33 (2019).
1. **Yahalomi, D. A.**, Schechter, P. L., and Wambsganss, J. "A Quadruply Lensed SN Ia: Gaining a Time-Delay…Losing a Standard Candle." MIT JUR (2017), arXiv:1711.07919.

Second & Third Author Publications

8. Nesvorný, D., **Yahalomi, D. A.**, et al. (2nd author) "ExoMOD II. A Statistical Model of Transit Timing Variations in Kepler Multi-Planet Systems." ApJ, 1009, 205 (2026).
7. Nesvorný, D., **Yahalomi, D. A.**, et al. (2nd author) "ExoMOD I. A Forward Model for the Orbital Architecture of Kepler Multi-Planet Systems." ApJ, 1008, 203 (2026).
6. Yang, E. et al. incl. **Yahalomi, D. A.** (3rd author) "A Massive Hot-Jupiter Companion that Disfavors Giant Planet Formation Beyond the Water-Ice Line." ApJL, 1004, L29 (2026).
5. Cassese, B. et al. incl. **Yahalomi, D. A.** (3rd author) "Quantifying the Contamination of TESS Ecliptic-plane Light Curves by Minor Planets." AJ, 171, 3 (2026).
4. Kipping, D. et al. incl. **Yahalomi, D. A.** (3rd author) "Near-circular orbits for planets with Earth-like sizes and insolations around M and K dwarf stars." Nature Astronomy, 9, 1007 (2025).
3. Kipping, D. et al. incl. **Yahalomi, D. A.** (3rd author) "Concerning the possible exomoons around Kepler-1625 b and Kepler-1708 b." Nature Astronomy, 9, 795 (2025).
2. Kipping, D. and **Yahalomi, D. A.** (2nd author) "A search for transit timing variations within the exomoon corridor using Kepler data." MNRAS, 518, 3 (2023).
1. Palatnick, S., Kipping, D., and **Yahalomi, D. A.** "Validation of HD 183579b Using Archival Radial Velocities: A Warm Neptune Orbiting a Bright Solar Analog." ApJL, 909, 1 (2021).

Independent Significant Contribution

5. Larsen, S. S. et al. **including Yahalomi D. A.** "TESS Photometry and Radial-velocity Analysis of the Sub-Neptune Exoplanet π Mensae c and the Wider π Mensae Planetary System." The Astronomical Journal, 172, 3 (2026).
4. Kipping, D. et al. incl. **Yahalomi, D. A.** "A JWST Transit of a Jupiter Analog: II. A Search for Exomoons." AJ, 172, 2 (2026).
3. Cassese, B. et al. **including Yahalomi D. A.** "A JWST Transit of a Jupiter Analog I: Constraints on the Oblateness of Kepler-167e." The Astronomical Journal, 171, 3 (2026).
2. Grunblatt, S. et al. **including Yahalomi D. A.** "Roman CCS White Paper: Adding Fields Hosting Globular Clusters To The Galactic Bulge Time Domain Survey." White Paper, arXiv:2306.10647 (2023).
1. Christian, S. et al. **including Yahalomi D. A.** "A Possible Alignment Between the Orbits of Planetary Systems and their Visual Binary Companions." The Astronomical Journal, 163, 5 (2022).

TESS Collaboration Papers

My authorship results from my contributions to mission planning, ground-based observing, and/or internal data analysis in the TESS collaboration. In all such instances, I provided substantive feedback on the manuscript.

18. Giacalone, S. et al. **including Yahalomi D. A.** "Validation of 13 Hot and Potentially Terrestrial TESS Planets." The Astronomical Journal, 163, 2 (2022).
17. Ikwut-Ukwa, M. et al. **including Yahalomi D. A.** "Two Massive Jupiters in Eccentric Orbits from the TESS Full Frame Images." The Astronomical Journal, 163, 1 (2022).
16. Scarsdale, N. et al. **including Yahalomi D. A.** "TESS-Keck Survey. V. Twin Sub-Neptunes Transiting the Nearby G Star HD 63935." The Astronomical Journal, 162, 5 (2021).
15. Teske, J. et al. **including Yahalomi D. A.** "The Magellan-TESS Survey. I. Survey Description and Midsurvey Results." The Astrophysical Journal Supplement Series, 256, 2 (2021).
14. Hoyer, S. et al. **including Yahalomi D. A.** "TOI-220 b: a warm sub-Neptune discovered by TESS." Monthly Notices of the Royal Astronomical Society, 505, 3 (2021).
13. Dong, J. et al. **including Yahalomi D. A.** "Warm Jupiters in TESS Full-Frame Images: A Catalog and Observed Eccentricity Distribution for Year 1." The Astrophysical Journal Supplement, 255, 1 (2021).
12. Guerrero, N. M. et al. **including Yahalomi D. A.** "The TESS Objects of Interest Catalog from the TESS Prime Mission." The Astrophysical Journal Supplement, 254, 2 (2021).
11. Rodriguez, J. E. et al. **including Yahalomi D. A.** "TESS Delivers Five New Hot Giant Planets Orbiting Bright Stars from the Full Frame Images." The Astronomical Journal, 161, 4 (2021).
10. Zhou, G. et al. **including Yahalomi D. A.** "Two Young Planetary Systems around Field Stars with Ages between 20 and 320 Myr from TESS." The Astronomical Journal, 161, 1 (2021).
9. Brahm, R. et al. **including Yahalomi D. A.** "TOI-481 b and TOI-892 b: Two Long-period Hot Jupiters from the Transiting Exoplanet Survey Satellite." The Astronomical Journal, 160, 5 (2020).
8. Beatty, T. G. et al. **including Yahalomi D. A.** "The TESS Phase Curve of KELT-1b Suggests a High Dayside Albedo." The Astronomical Journal, 160, 211 (2020).
7. Ikwut-Ukwa, M. et al. **including Yahalomi D. A.** "The K2 & TESS Synergy I: Updated Ephemerides and Parameters for K2-114, K2-167, K2-237, & K2-261." The Astronomical Journal, 160, 209 (2020).
6. Wong, I. et al. **including Yahalomi D. A.** "Systematic Phase Curve Study of Known Transiting Systems from Year 1 of the TESS Mission." The Astronomical Journal, 160, 155 (2020).
5. Mireles, I. et al. **including Yahalomi D. A.** "TOI 694 b and TIC 220568520 b: Two Low-Mass Companions Near the Hydrogen Burning Mass Limit Orbiting Sun-like Stars." The Astronomical Journal, 160, 133 (2020).
4. Wong, I., et al. **including Yahalomi, D. A.** "Exploring the atmospheric dynamics of the extreme ultra-hot Jupiter KELT-9b using TESS photometry." The Astronomical Journal, 160, 88 (2020).
3. Dragomir, D. et al. **including Yahalomi D. A.** "Securing the Legacy of TESS through the Care and Maintenance of TESS Planet Ephemerides." The Astronomical Journal, 159, 219 (2020).
2. Diaz, M. R. et al. **including Yahalomi D. A.** "TOI-132 b: A short-period planet in the Neptune desert transiting a V=11.3 G-type star." Monthly Notices of the Royal Astronomical Society, 493, 973 (2020).
1. Rodriguez, J., et al. **including Yahalomi, D. A.** "An Eccentric Massive Jupiter Orbiting a Sub-Giant on a 9.5 Day Period Discovered in the TESS Full Frame Images." The Astronomical Journal, 157, 191 (2019).

Advising

Primary advisor to 2 undergraduates and 12 high school students; co-advisor to 4 undergraduates.

As Primary Advisor — Undergraduates

- 2026–present **Aditya Peddi** (MIT, UROP). Unraveling planet–planet transit timing variations.
- 2023–2024 **Yassine Abaakil** (Columbia). Masses and radii of the white dwarfs in four *Kepler* self-lensing binaries.

As Primary Advisor — High School Students (Columbia STAR; Harvard–MIT SRMP)

- 2025–2026 **Tayeba Kafil, Meita Konate, Maxwell Lally** (Columbia Secondary School, STAR). Can life's fingerprints survive on Mars?
- 2024–2025 **Nasiah Anderson** (now Lafayette College), **Mark Driker** (now Northwestern), **Kokoro Onuma** (Columbia Secondary School, STAR). Micrometeoroid impact rates for an Artemis-era lunar base.
- 2022–2023 **Farai Sundai** (now Northwestern), **Jiajing Liu** (now Minnesota), **Lila Valaskovic** (now Colgate) (Cambridge Rindge & Latin, SRMP). Transit timing variations in planetary hierarchical triples.
- 2020–2021 **Mohammed Sakib** (now Harvard), **Tsion Tedla** (now Boston University), **Victoria Chen** (now Toronto) (Cambridge Rindge & Latin, SRMP). Modeling the radial velocities of four *Kepler* self-lensing binaries.

As Co-Advisor — Undergraduates

- 2025–2026 **John Moynihan** (Columbia). Profile-likelihood analysis of transit timing variations.
- Summer 2022 **Andrew Zhang, Avishi Poddar, Madison Li** (Columbia). "Democratically" detrending *TESS* M dwarfs.

Outreach

Columbia Student Training in Astronomy Research (STAR) Program — *New York, NY*

Advisory Director	*2026 – present*
Founder & Director	*2024 – 2026*
Research Mentor	*2024 – 2026*

Harvard-MIT Science Research Mentoring Program (SRMP) — *Cambridge, MA*

Co-Director	*2020 – 2023, 2026 – present*
Research Mentor	*2020 – 2021, 2022 – 2023*
Head of Observing	*2018 – 2020*

Professional Activities

- **Journal Referee**, AAS Journals (2023 – present), Universe (2023 – present).
- **Organizing Committee**, Gaia Follow-up Observing Program (2026 – present).
- **Organizing Committee**, NASA ExoExplorers Alumni Mentorship Program (2025 – present).
- **Organizing Committee**, *Gaia* Warm-up at MIT (November 2026).
- **Organizing Committee**, Planets in the Galactic Context Workshop, CCA (October 2026).
- **Organizing Co-Chair**, Tooling up for *Gaia* Workshop, CCA (April 2026).
- **Graduate Student Representative**, Columbia Astronomy Faculty Meetings: 2024-2025.
- **Graduate Student Representative**, Columbia Astronomy Faculty Search Committee: 2024.
- **Graduate Student Faculty Liaison**, Columbia Astronomy Department: 2022-2024.
- **Graduate Student Representative**, Columbia Astronomy Faculty Search Committee: 2022.
- **Member**, American Astronomical Society (2018 – present).
- **Member**, TESS Follow-up Observing Program (2018 – present).
- **Associate Member**, Sigma Xi (2020 – present).
- **Science Alliance Member**, New York Academy of Sciences (2020 – present).

Observing Proposals

As PI

3. **Yahalomi, D. A.** et al. "Photometric Confirmation and Ephemeris Refinement of TESS Planet Candidates.", MDM McGraw-Hill 1.3m Telescope 2023b, 8 nights.
2. **Yahalomi, D. A.** et al. "Photometric Confirmation and Ephemeris Refinement of TESS Planet Candidates.", MDM McGraw-Hill 1.3m Telescope 2022a, 4 nights.

1. **Yahalomi, D. A.** et al. "Photometric Confirmation and Ephemeris Refinement of TESS Planet Candidates.", MDM McGraw-Hill 1.3m Telescope 2022b, 5 nights.

As Co-I or Collaborator

6. Cassese, B. et al. **including Yahalomi D. A.** "Fluffy or Flashy? Resolving the Mystery of the Coldest-Known Super Puff.", JWST Cycle 5, General Observer Proposal (GO 11253).
5. Pooley, D. A. et al. **including Yahalomi D. A.** "Nano-arcsecond Tomography of the Central Regions of the Quasar in SDSS J0924+0219.", Chandra Cycle 26, Large Target of Opportunity Proposal.
4. Cassese, B. and **Yahalomi, D. A.** "Attempted Recovery of a Distant Trans-Neptunian Object." MDM Hiltner 2.4m Telescope 2022b, 5 nights.
3. Pooley, D. A. et al. **including Yahalomi D. A.** "Nano-arcsecond Tomography of the Central Regions of the Quasar in SDSS J0924+0219.", Chandra Cycle 24, Large Target of Opportunity Proposal.
2. Pooley, D. A. et al. **including Yahalomi D. A.** "Nano-arcsecond Tomography of the Central Regions of the Quasar in SDSS J0924+0219.", Chandra Cycle 23, Large Target of Opportunity Proposal.
1. Angus, R. et al. **including Yahalomi D. A.** "Measuring long rotation periods from TESS light curves.", NASA TESS Guest Investigator program, Cycle 3, large program.

Selected Talks

- **Planets in the Galactic Context** (*New York, NY*), October 2026.
- **Multiple Planet Systems** (*Sofia, Bulgaria*), September 2026.
- **AAS DDA 57th Meeting** (*Chicago, IL*), June 2026.
- **MIT TESS Science Talk** (*Cambridge, MA*), April 2026.
- **MIT MKI Journal Club Talk** (*Cambridge, MA*), March 2026.
- **Preparing for Gaia's Exoplanet Bonanza** (*Ringberg, Germany*), March 2026.
- **Boston Area Planetary Science Meeting** (*Boston, MA*), January 2026.
- **A Dynamical Synthesis of Planetary Systems Conference** (*New Haven, CT*), January 2026.
- **International Conference on Exoplanets and Planet Formation** (*Shanghai, China*), December 2025.
- **CCA Coffee Talk** (*New York, NY*), November 2025.
- **New York Area Exoplanets Meeting** (*New York, NY*), May 2025.
- **Machine Learning in Planetary System Dynamics CCA Workshop** (*New York, NY*), April 2025.
- **Weizmann Institute of Science Seminar** (*Rehovot, Israel*), February 2025.
- **Circumplanetary Disks and Satellite Formation III** (*Kyoto, Japan*), January 2025.
- **AAS 245th Meeting** (*National Harbor, MD*), January 2025.
- **Yale Exoplanet & Stars Seminar** (*New Haven, CT*), November 2024.
- **New York Area Exoplanets Meeting** (*New York, NY*), May 2024.
- **Princeton Exoplanet Lunch Talk** (*Princeton, NJ*), May 2024.
- **Bard College Physics Colloquium** (*Annandale-On-Hudson, NY*), April 2024.
- **Columbia Astronomy Public Talk** (*New York, NY*), April 2024.
- **NASA ExoExplorers Talk** (*Zoom*), April 2024.
- **MIT TESS Science Talk** (*Cambridge, MA*), April 2024.
- **Harvard CfA Exoplanet Pizza Lunch** (*Cambridge, MA*), April 2024.
- **Extreme Solar Systems V** (*Christchurch, New Zealand*), February 2024.
- **Weizmann Institute of Science Seminar** (*virtual*), February 2024.
- **University of Colorado APS Seminar** (*Boulder, CO*), January 2024.
- **AAS 243rd Meeting** (*New Orleans, LA*), January 2024.
- **SwRI Boulder Colloquium** (*Boulder, CO*), January 2023.
- **Harvard CfA Stars and Planets Seminar** (*Cambridge, MA*), May 2021.
- **AAS 237th Meeting** (*virtual*), January 2021

- **Princeton Exoplanet Lunch Meeting** (*Princeton, NJ*), November 2019.
- **Harvard CfA Exoplanet Pizza Lunch** (*Cambridge, MA*), November 2019.
- **AAS 233rd Meeting** (*Seattle, WA*), January 2019.
- **MIT Cosmology Undergraduate Workshop** (*Cambridge, MA*), August 2017.
- **Columbia Nevis Laboratory Talk** (*New York, NY*), June 2017.
- **Manhattan Microlensing Conference** (*New York, NY*), June 2017.

Teaching

Columbia Astronomy Department — *New York, NY*

- **Teaching Assistant:** Astrostatistics ... Graduate & Undergraduate Course — *Spring 2023*
- **Instructor:** Observing Labs ... Undergraduate Course — *2021 – 2022*
- **Teaching Assistant:** Earth, Moon, and Planets ... Undergraduate Course — *Summer 2021*
- **Teaching Assistant:** Stars and Atoms ... Undergraduate Course — *Spring 2021*
- **Teaching Assistant:** Another Earth ... Undergraduate Course — *Fall 2020*

MIT Physics Department — *Cambridge, MA*

- **Teaching Assistant:** Intro to Mechanics Review (8.01R) — *Jan 2015*